

# ETP story-formalization benchmark

Stories generated from Equational Theories Project implications (`storyform.py`), formalized back by each model over OpenRouter, and graded syntactically (`checkform.py`). Correct = exact, or an accepted symmetry (sides swapped, or both equations uniformly dualized).

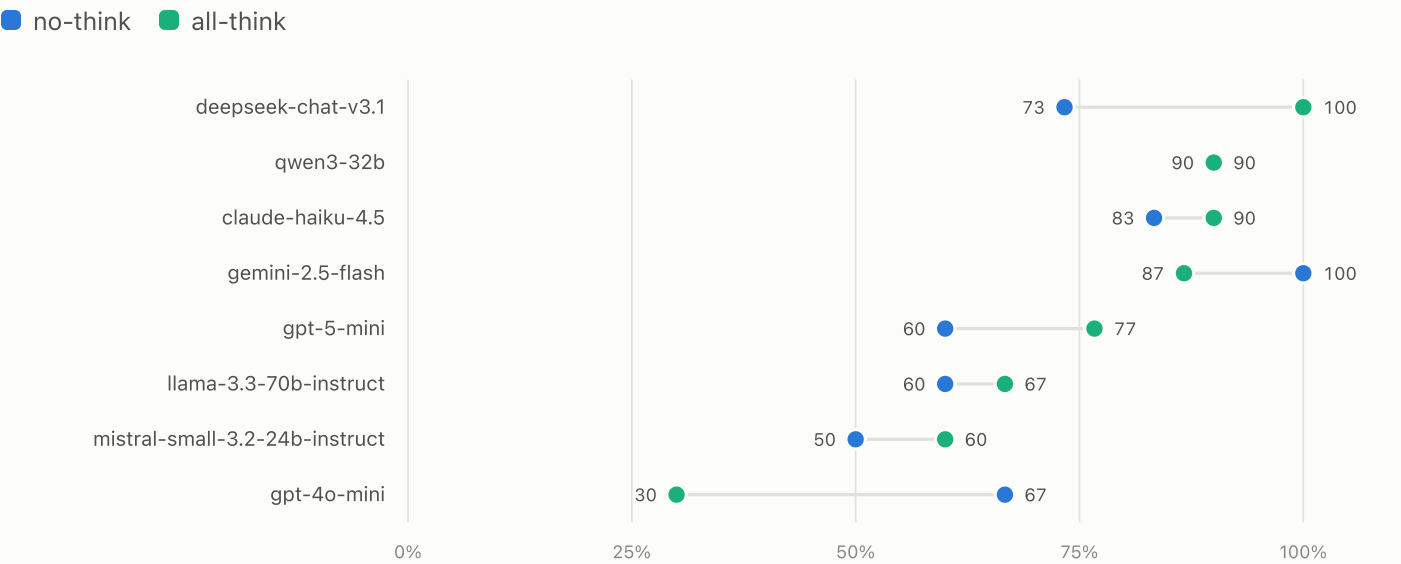
experiments/03-literal-reasoning-and-complexity/runs/run-s0-n30-think-off · experiments/03-literal-reasoning-and-complexity/runs/run-s0-n30-think-on · experiments/03-literal-reasoning-and-complexity/runs/run-strat5-s0-think-off · experiments/03-literal-reasoning-and-complexity/runs/run-strat5-s0-think-on

|      |        |         |              |                 |
|------|--------|---------|--------------|-----------------|
| Runs | Models | Stories | Graded calls | Overall correct |
| 4    | 8      | 70      | 1,120        | 84%             |

## 30 uniformly sampled pairs · seed 0 · no-think vs all-think

FIGURE 1 **Correct rate by model: no-think vs all-think**

Same 30 stories under both regimes; each row connects a model's cold-pass rate to its deliberate rate.

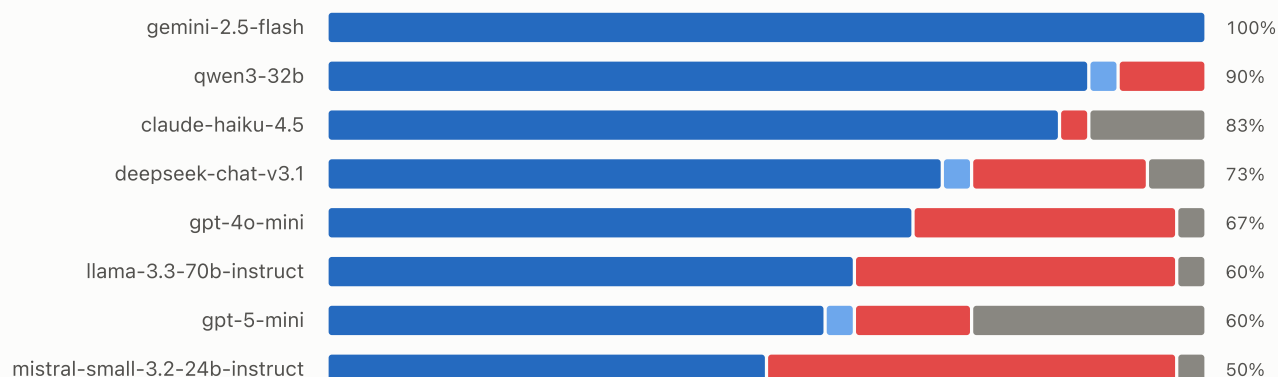


► Data: correct rate per model and regime

FIGURE 2 **Verdict composition — no-think regime**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable

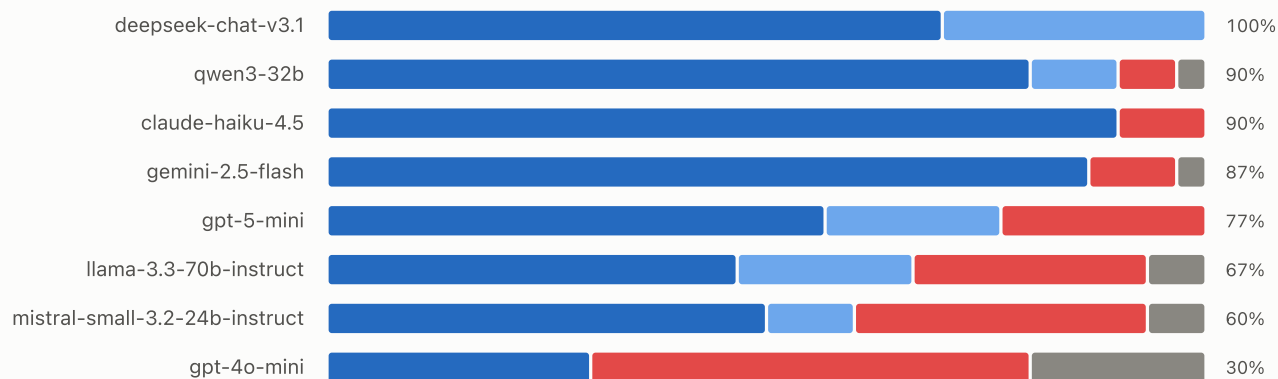


► Data: verdict counts per model

FIGURE 3 **Verdict composition — all-think regime**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).

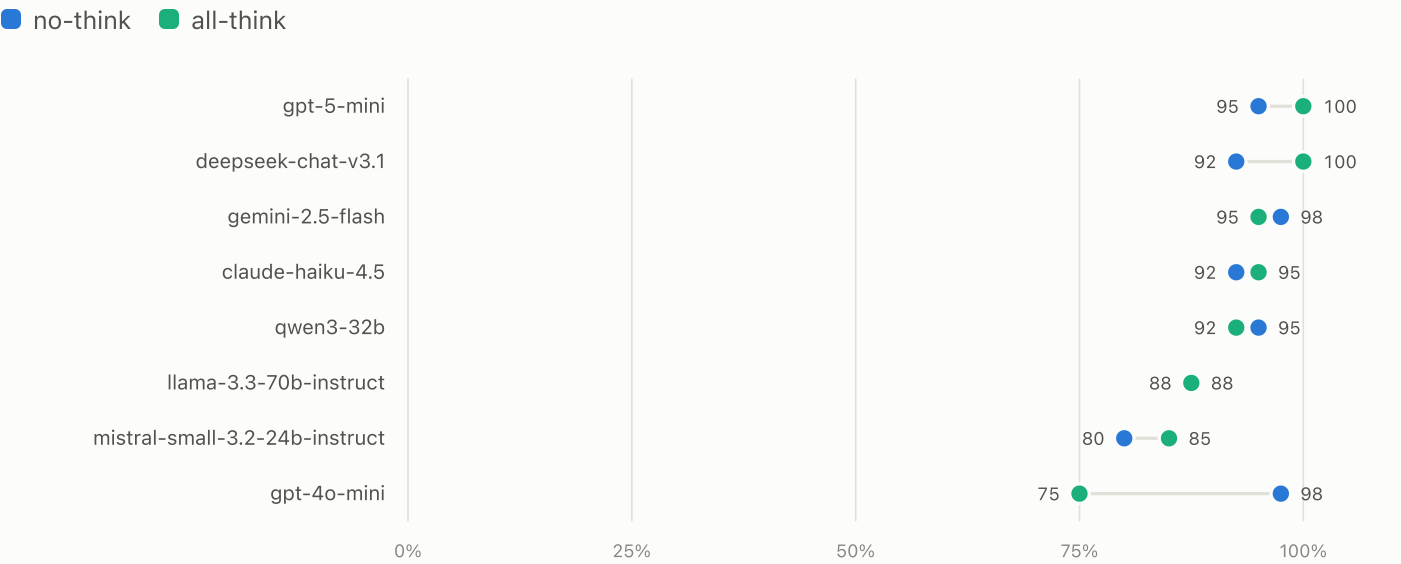
■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 4 **Correct rate by model: no-think vs all-think**

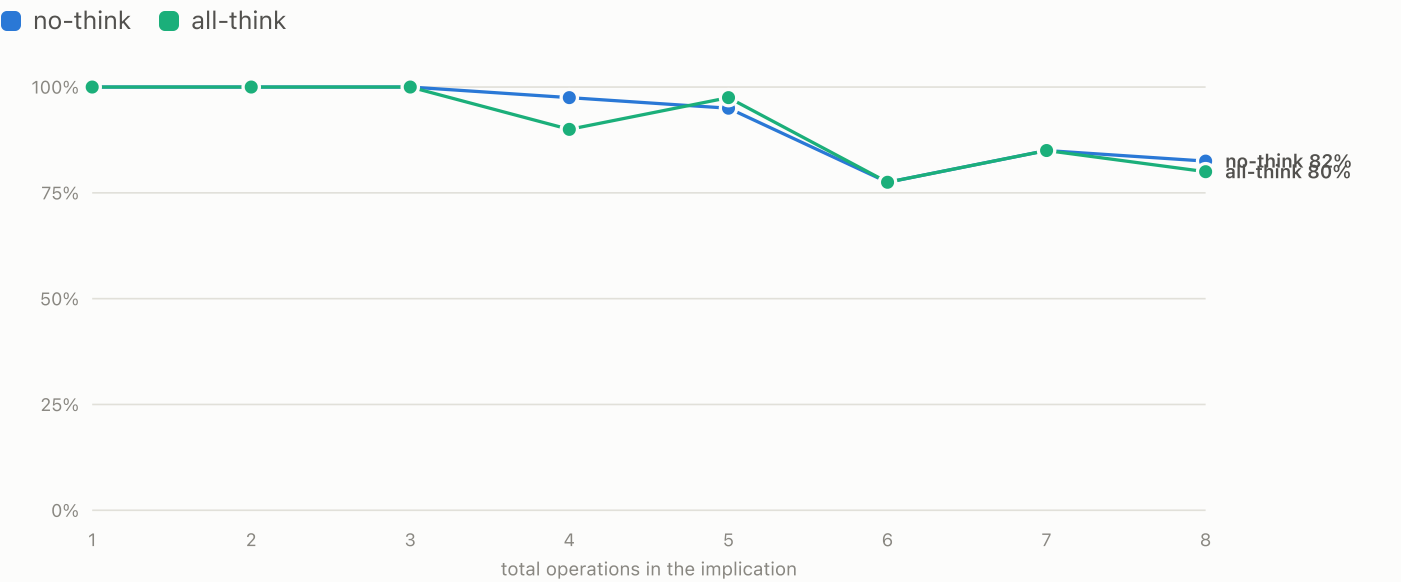
Same 40 stories under both regimes; each row connects a model's cold-pass rate to its deliberate rate.



► Data: correct rate per model and regime

FIGURE 5 **Accuracy vs implication complexity**

Correct rate pooled over all models, by the implication's total operation count (both equations combined).



► Data: pooled correct rate per operation-count bin

FIGURE 6 **Verdict composition — no-think regime**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable

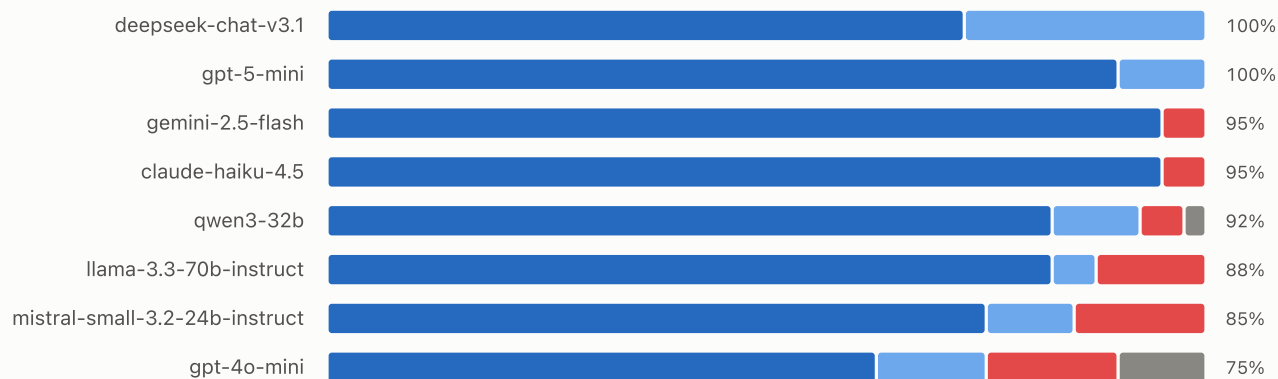


► Data: verdict counts per model

FIGURE 7 **Verdict composition — all-think regime**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model